Computer Networking: Definition, Types of Networking, and Network Topology

1. What Is Networking?

Networking is the process of connecting two or more computers or devices so that they can communicate and share data, resources, and services with each other.

In simple words, networking means connecting devices so they can exchange information.

For example, when two computers are connected to the same network, they can share files, use the same printer, or access the internet through a common router.



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Real-life examples of networking

- Computer lab: Multiple computers are connected to share files and a printer.
- Office: Employees access shared files and company servers.
- Home: Mobile phones, laptops, smart TVs, and printers connect to the same Wi-Fi router.
- Internet: Millions of networks connect worldwide to exchange information.
- Bank: Computers communicate with servers to process transactions.

Main components of networking



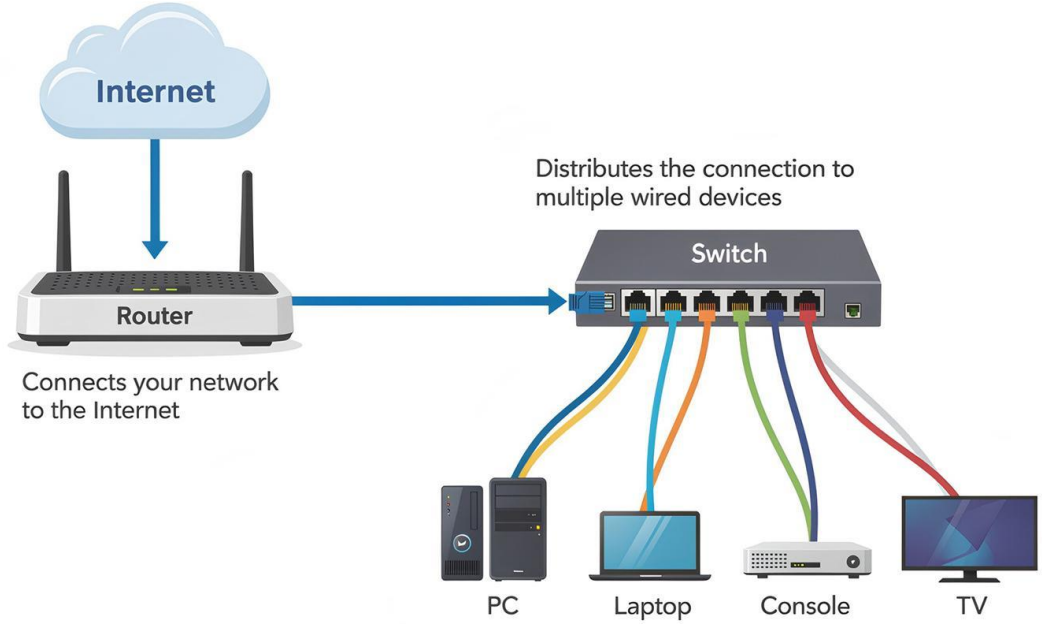
1. Computer / Client

A device that sends or receives data, such as a desktop computer, laptop, or smartphone.



2. Server

A computer that provides services or resources to other devices, such as files, websites, and applications.



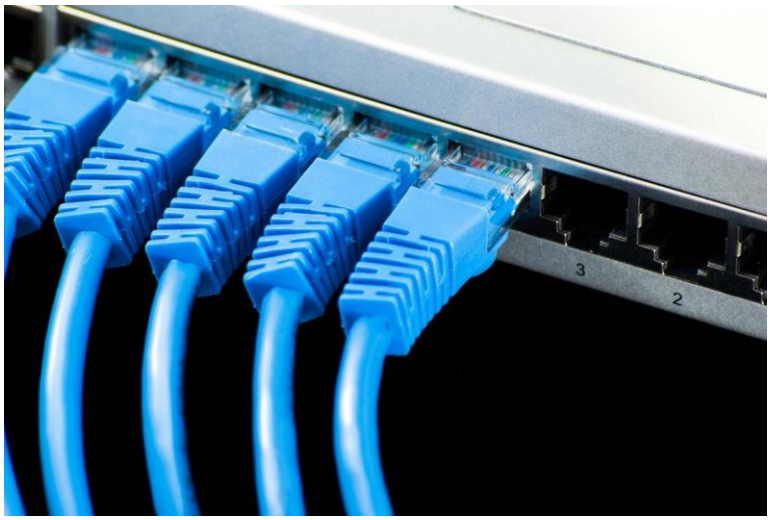
3. Router and Switch

A switch connects devices within a local network. A router forwards traffic between different networks.



4. Transmission Media

The medium through which data travels, such as Ethernet cables, fiber-optic cables, or wireless radio signals.



5. Network Interface Card (NIC)

Hardware that enables a device to connect to a network through Ethernet or Wi-Fi.

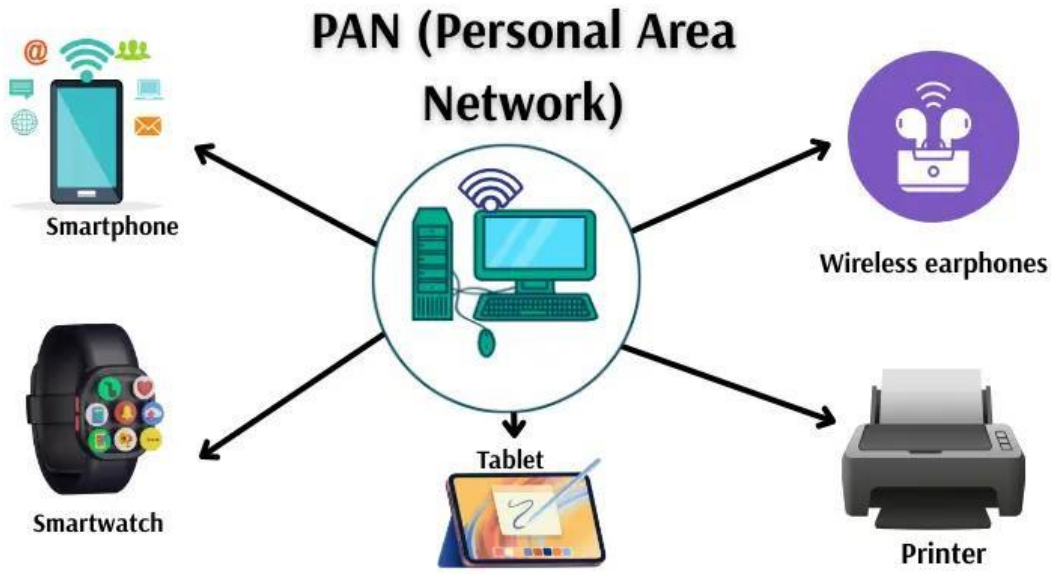
Advantages of networking

1. Resource sharing: Share printers, files, storage, and internet connections.
2. Communication: Exchange information through email, messaging, and video calls.
3. Centralized management: Manage users and files from a central server.
4. Data sharing: Transfer information between connected devices.
5. Cost reduction: Multiple users can share the same resources.
6. Remote access: Access authorized systems from other locations.
7. Security management: Apply centralized access controls and monitoring.

2. Types of Networking

Networks are commonly classified according to their geographical coverage, purpose, and connection method.

The following are the major types of networks.



1. PAN — Personal Area Network

Full form: Personal Area Network

A PAN connects personal devices over a very short distance, usually around one person.

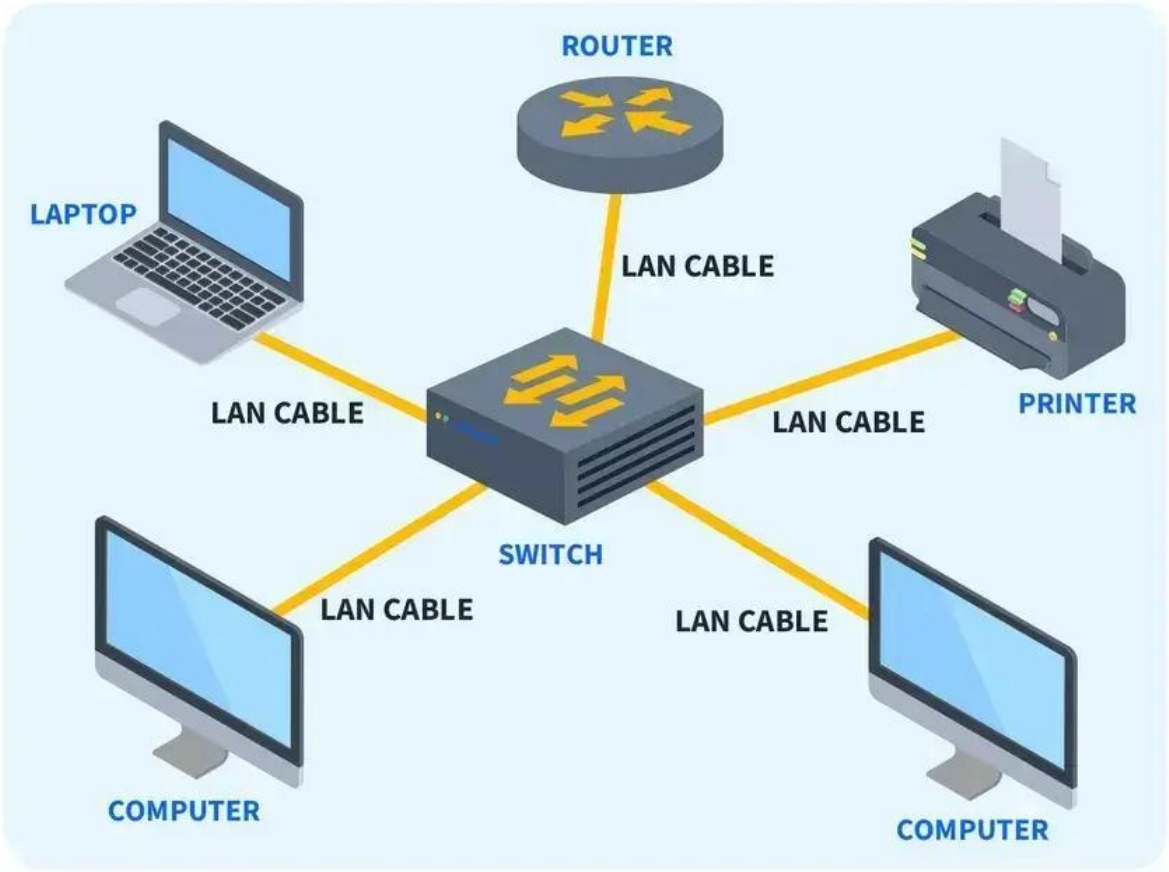
Examples:

- Connecting wireless earbuds to a mobile phone.
- Connecting a smartwatch to a smartphone.
- Sharing files between nearby devices using Bluetooth.

Coverage: Usually a few meters.

Advantages: Easy to set up, portable, and convenient.

Limitation: Short range and limited capacity compared with larger networks.



2. LAN — Local Area Network

Full form: Local Area Network

A LAN connects devices within a limited geographical area, such as a home, classroom, office, or building.

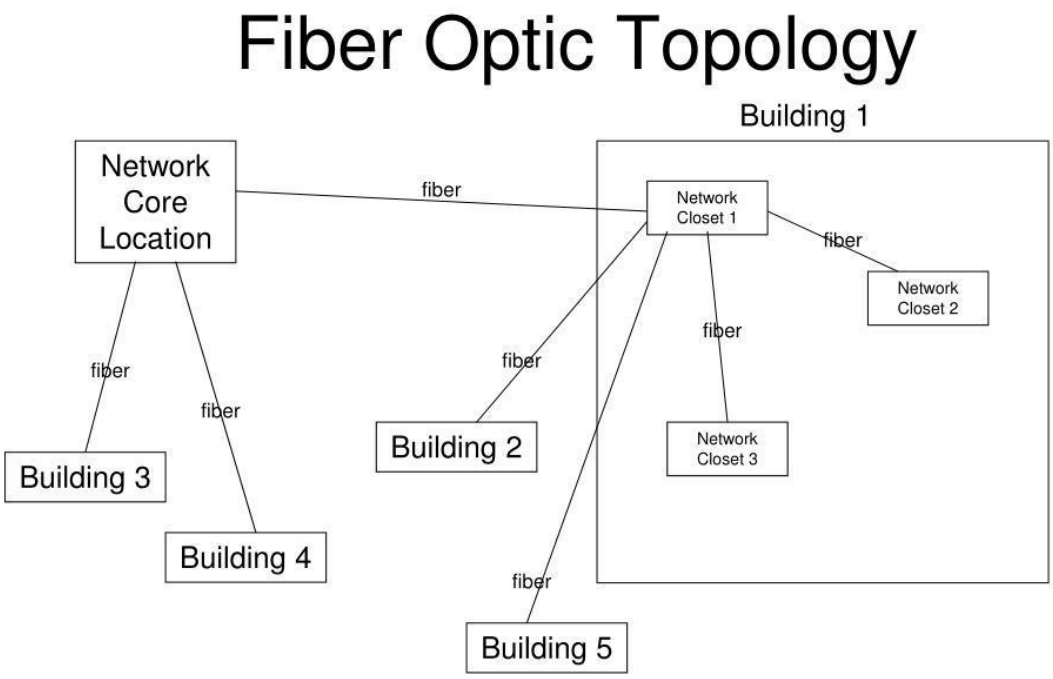
Examples:

- A computer laboratory in an institute.
- Computers connected to an office switch.
- A home network using a Wi-Fi router.

Coverage: A room, building, or nearby buildings.

Advantages: High speed, resource sharing, and relatively low operating cost.

Limitation: Coverage is limited to the local site unless connected to other networks.



3. CAN — Campus Area Network

Full form: Campus Area Network

A CAN connects multiple LANs within a campus or a group of nearby buildings.

Examples:

- A university connecting its library, laboratories, and administrative offices.
- A company connecting multiple buildings at one campus.

Coverage: A campus or a group of nearby buildings.

Advantages: Centralized communication and resource sharing across the campus.

Limitation: Requires planning, network equipment, and maintenance.



4. MAN — Metropolitan Area Network

Full form: Metropolitan Area Network

A MAN connects multiple LANs or smaller networks across a city or metropolitan area.

Examples:

- Connecting company offices across Gandhinagar.
- A city-wide institutional network.
- A service provider's metropolitan fiber network.

Coverage: A city or metropolitan region.

Advantages: Connects locations across a city and supports centralized services.

Limitation: More complex and expensive than a typical LAN.



5. WAN — Wide Area Network

Full form: Wide Area Network

A WAN connects networks over large geographical distances, such as between cities, states, or countries.

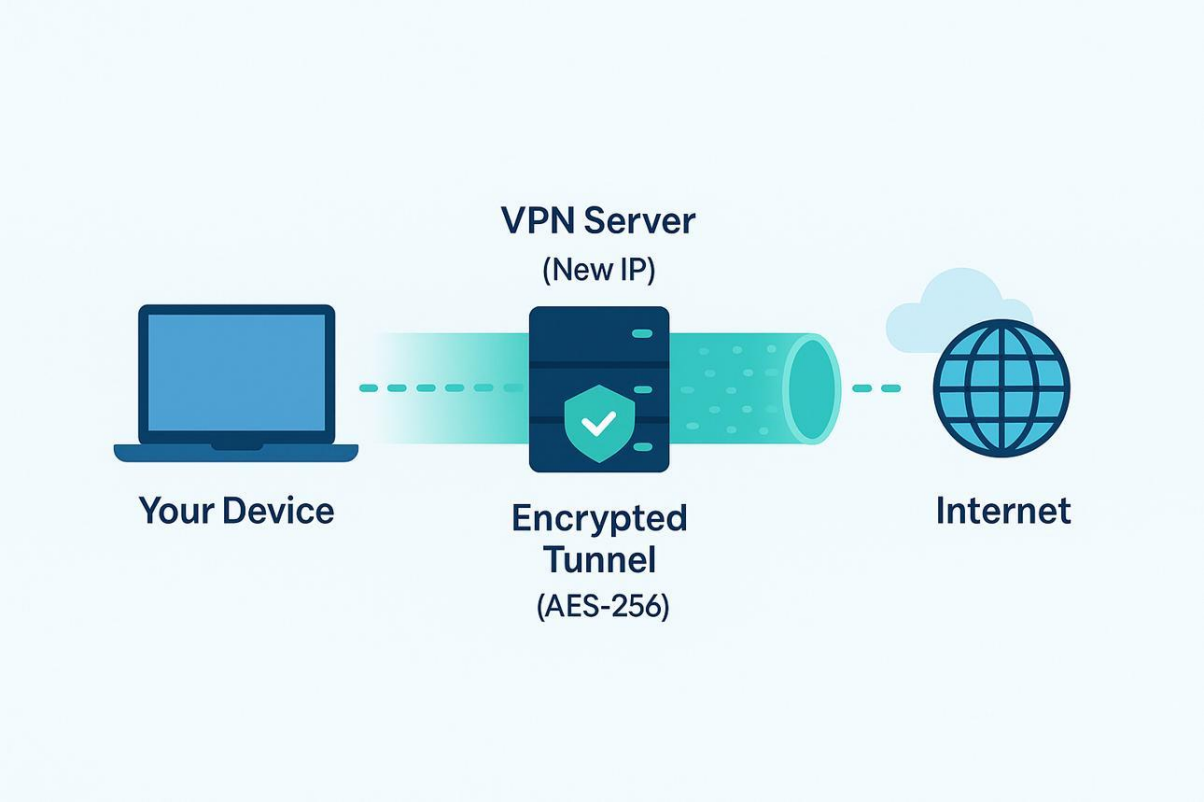
Examples:

- A bank connecting branches across India.
- A company connecting international offices.
- The internet, which interconnects networks worldwide.

Coverage: Large geographical areas, potentially worldwide.

Advantages: Long-distance communication and connectivity between remote offices.

Limitation: Can be expensive and requires careful management of performance and security.



6. VPN — Virtual Private Network

Full form: Virtual Private Network

A VPN creates an encrypted connection over an existing network, allowing authorized users to access resources securely.

Examples:

- An employee securely accessing the office network from home.
- A remote worker connecting to company systems.

Coverage: Depends on the underlying internet or network connection.

Advantages: Helps protect traffic in transit and enables secure remote access.

Limitation: Does not guarantee complete anonymity or protect against every cyber threat.

Important: A VPN is a virtual networking technology, rather than a geographical network category like LAN or WAN.

Comparison of network types

Network	Full form	Typical coverage	Example
PAN	Personal Area Network	Around one person	Bluetooth devices
LAN	Local Area Network	Room/building	Computer lab
CAN	Campus Area Network	Campus	University
MAN	Metropolitan Area Network	City	City-wide network
WAN	Wide Area Network	Countries/worldwide	Bank branches
VPN	Virtual Private Network	Depends on underlying network	Secure remote access

Note: Coverage ranges are approximate. The actual range depends on the technology and network design. Other important network types include WLAN (Wireless LAN), SAN (Storage Area Network), and VLAN (Virtual LAN).

3. What Is Network Topology?

Network topology is the physical or logical arrangement of devices and connections in a computer network.

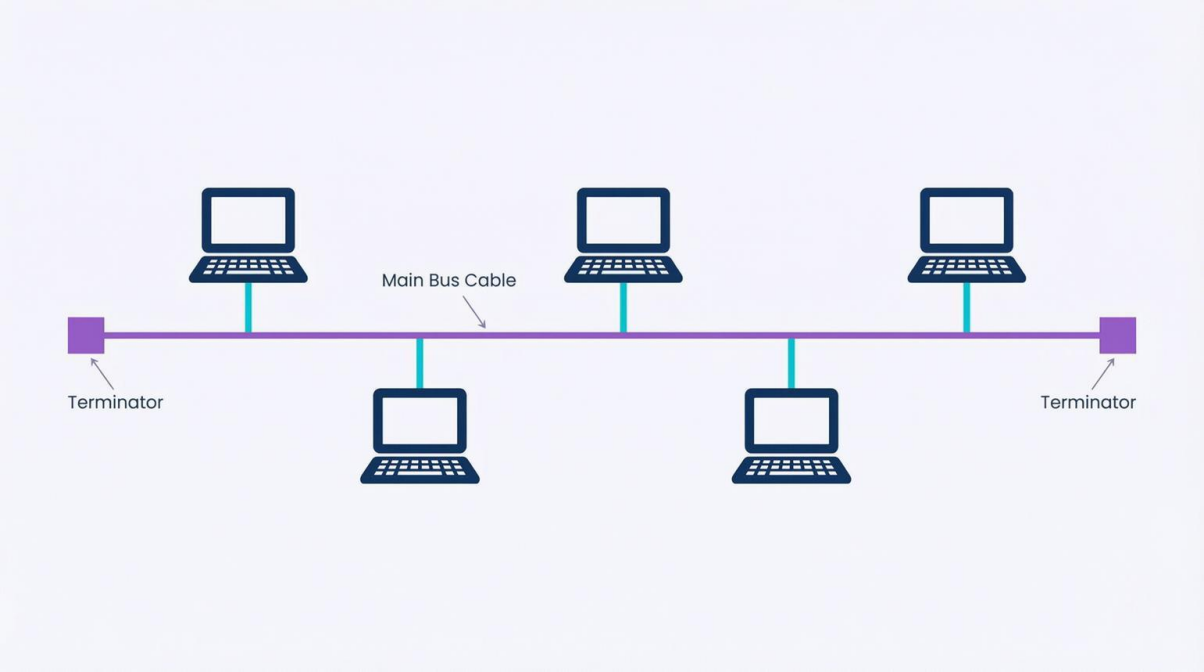
In simple words, topology describes how computers, switches, servers, and other network devices are connected to each other.

For example, computers can connect to one central switch, form a circular path, or connect directly to a shared backbone cable. Each arrangement is a different topology.

There are several common network topologies.

4. Types of Network Topology

1. Bus Topology



In bus topology, all devices connect to one main cable called the backbone.

How it works: Data travels along the shared cable, and devices receive the signals.

Advantages

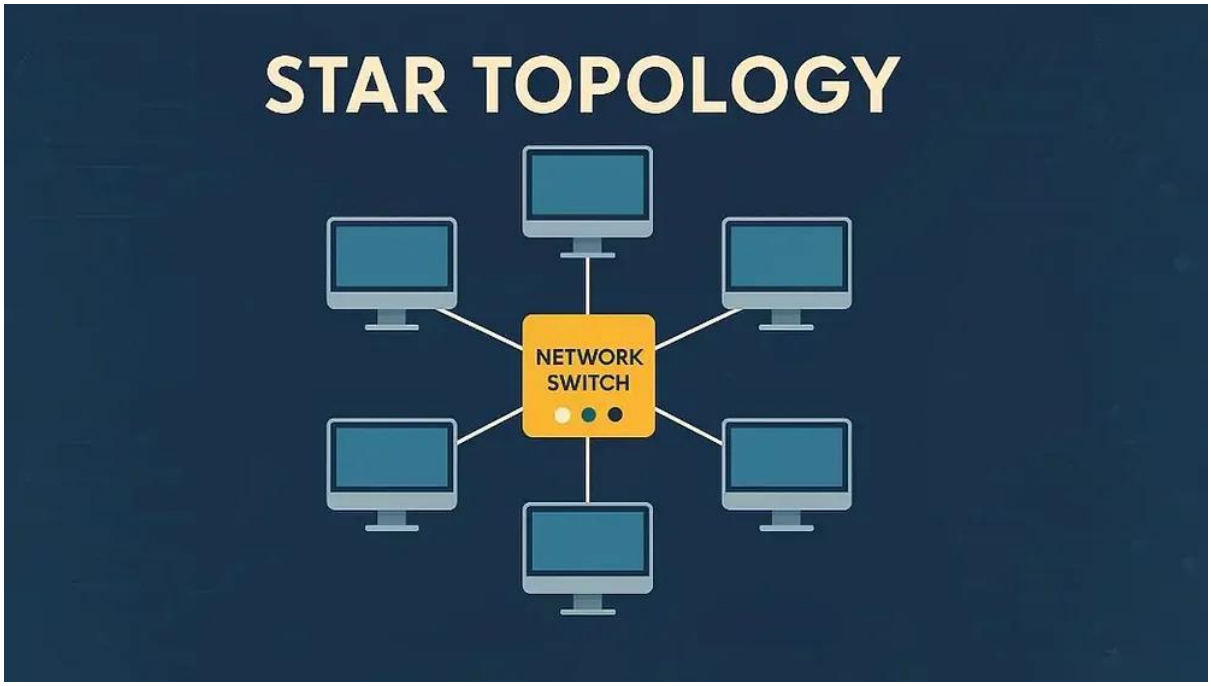
- Simple design.
- Uses less cable than some other topologies.
- Can be inexpensive for small, older networks.

Disadvantages

- If the main backbone cable fails, the network may stop working.
- Troubleshooting can be difficult.
- Performance can decrease as network traffic increases.

Example: Older Ethernet networks that used coaxial backbone cables.

2. Star Topology



In star topology, every device connects to a central device, usually a switch.

How it works: When one computer sends data to another, the switch forwards the data to the appropriate destination.

Advantages

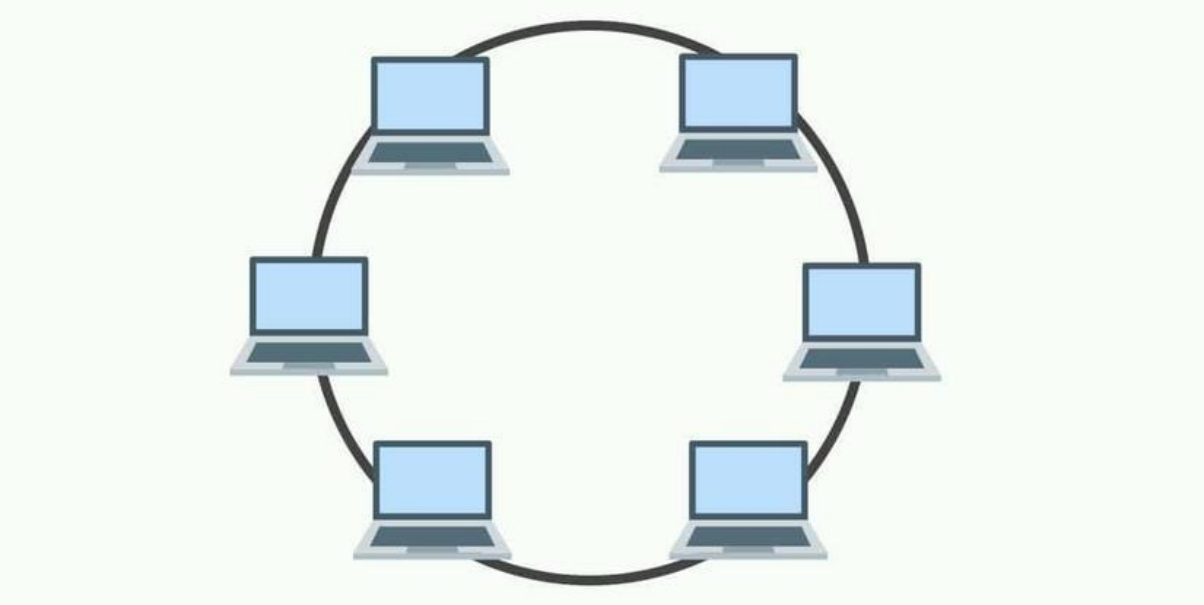
- Easy to install and manage.
- Easy to add or remove devices.
- Failure of one endpoint cable usually affects only that device.
- Faults are relatively easy to identify.

Disadvantages

- If the central switch fails, connected devices may lose network connectivity.
- Requires more cabling than a traditional bus topology.

Example: Modern office networks and computer laboratories.

3. Ring Topology



In ring topology, each device connects to two neighboring devices, creating a circular path.

How it works: Data travels around the ring according to the network's communication method.

Advantages

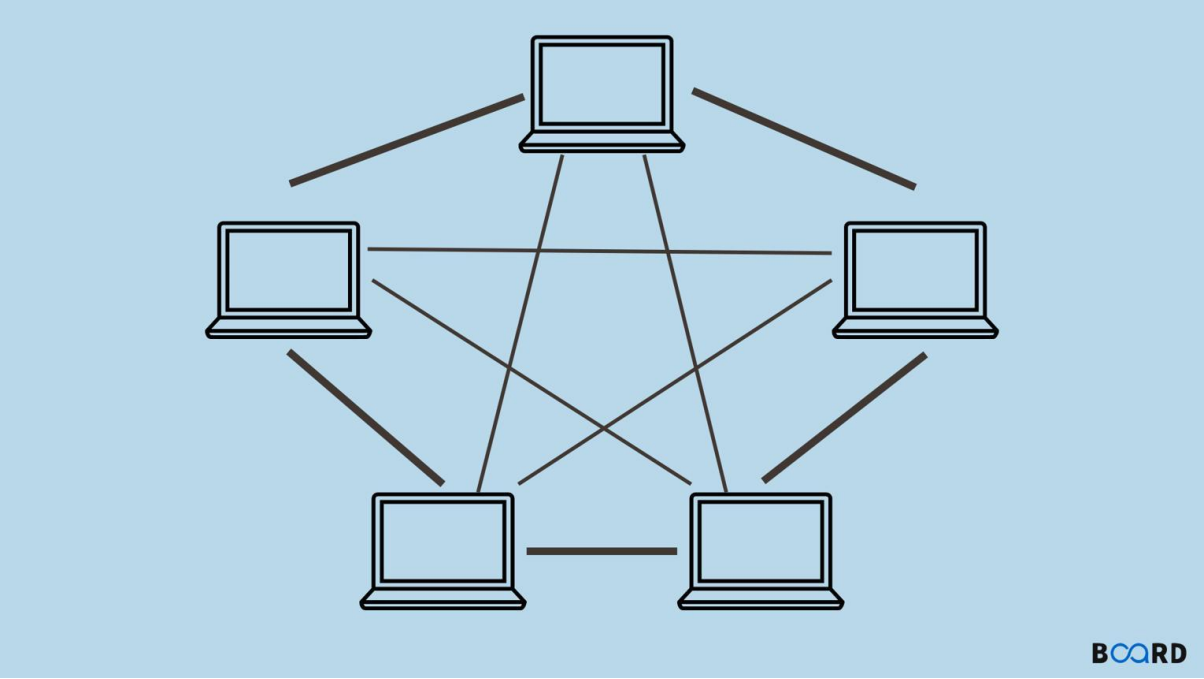
- Provides an organized communication path.
- Some ring systems can provide predictable performance under specific conditions.

Disadvantages

- A single link or device failure can disrupt a basic, unprotected ring.
- Troubleshooting and expansion can be difficult.

Example: Traditional Token Ring networks and certain ring-based fiber systems.

4. Mesh Topology



In mesh topology, devices are connected through multiple paths.

There are two main types:

- Full mesh: Every device connects directly to every other device.
- Partial mesh: Only selected devices have multiple direct connections.

Advantages

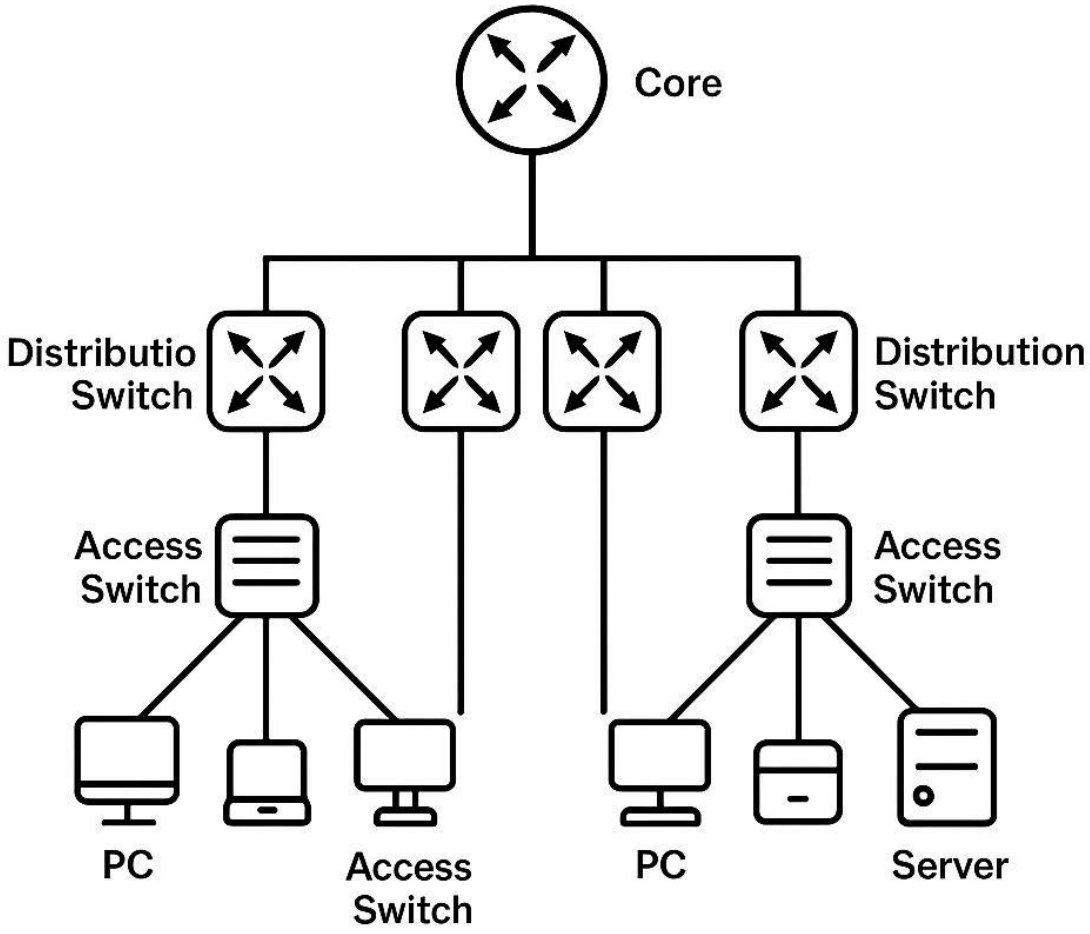
- Multiple paths can improve reliability.
- Traffic may be rerouted if a link fails.
- Useful where availability is important.

Disadvantages

- Full mesh requires many links.
- Installation and maintenance can be expensive.
- Network configuration can be complex.

Example: Wireless mesh networks, resilient backbone networks, and selected router interconnections.

5. Tree Topology



Tree topology organizes devices in a hierarchy, similar to the branches of a tree. It often combines multiple star networks.

How it works: A central or higher-level device connects to lower-level switches, which connect to end devices.

Advantages

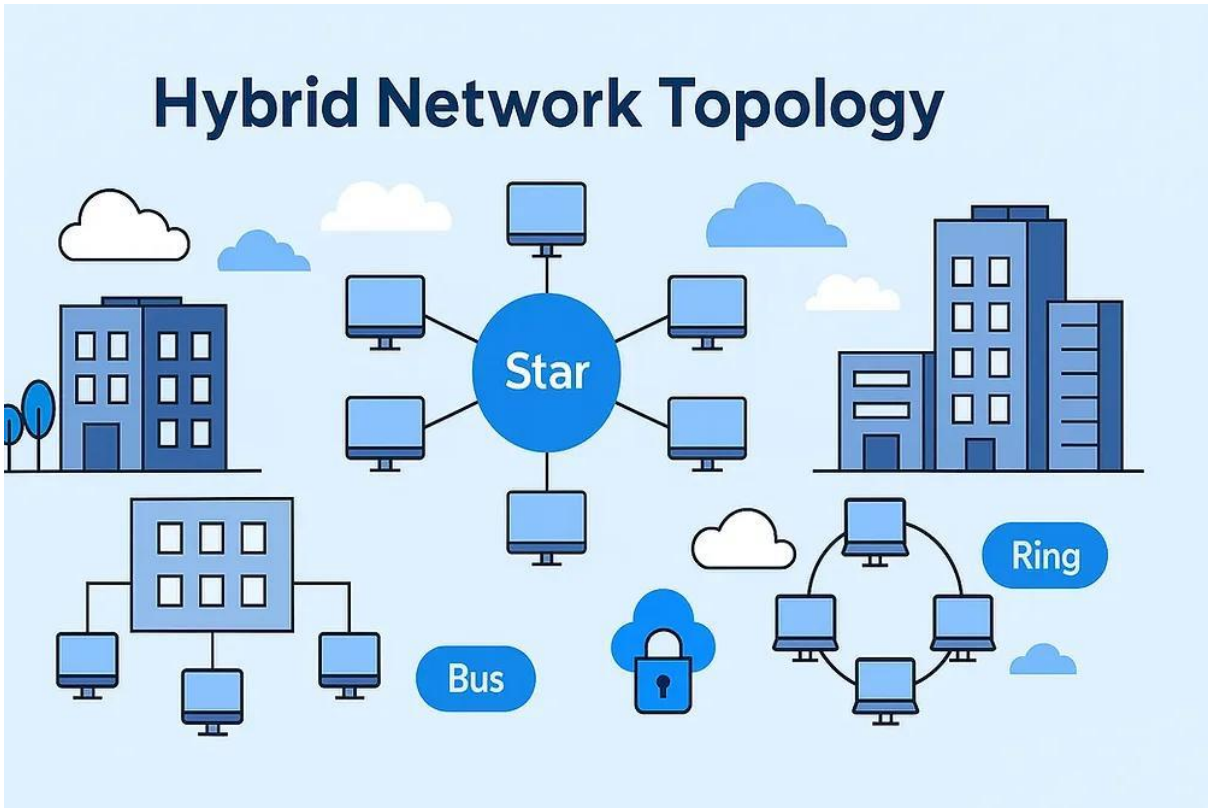
- Suitable for large networks.
- Easier to organize departments or buildings.
- Can be expanded by adding branches.

Disadvantages

- Failure of a major upstream device or link can affect an entire branch.
- Requires planning and additional networking equipment.

Example: A college network with a core switch, department switches, and classroom computers.

6. Hybrid Topology



Hybrid topology combines two or more topology types in one network.

How it works: Different sections use different arrangements and are connected together.

Advantages

- Flexible design.
- Can scale to meet different requirements.
- Allows different departments or buildings to use suitable topologies.

Disadvantages

- Design and troubleshooting can be complicated.
- Equipment and maintenance costs may be higher.

Example: An organization using star topology in each office and redundant links between buildings.

7. Point-to-Point Topology



Point-to-point topology creates a direct connection between two devices.

How it works: The two endpoints communicate over a dedicated or logically direct link.

Advantages

- Simple to configure.
- Direct communication between endpoints.
- Easy to understand and troubleshoot.

Disadvantages

- Connects only two endpoints on that link.
- A link failure interrupts communication over that connection unless an alternative path exists.

Example: A direct link between two routers or a dedicated connection between two sites.

5. Network Topology Comparison

Topology	Structure	Main advantage	Main disadvantage
Bus	One shared backbone	Less cabling in basic designs	Backbone failure can disrupt network
Star	Central switch or hub	Easy to manage	Central device is a dependency
Ring	Circular connection	Organized communication path	Basic ring may fail if a link breaks
Mesh	Multiple interconnected paths	High path redundancy	Expensive and complex
Tree	Hierarchical branches	Easy to expand systematically	Upstream failures can affect branches
Hybrid	Combination of topologies	Flexible design	Complex maintenance
Point-to-point	Direct link between two devices	Simple communication	Link failure breaks that connection